

Squared or Cubed? — Measurement Unit Detective

Directions:

- Every answer needs a *number* and the *right unit*. Flat space takes square units; filled space takes cubic units.
- Some problems show another student's work with a mistake in it. Find the mistake, then fix it.

1. A sticker is a rectangle 7 cm long and 4 cm wide. What is its area?

Answer: _____ cm^2

2. A small box is packed full of centimeter cubes (each cube is one centimeter on every side). Each layer holds 6 cubes, and the box holds 4 layers. What is the volume of the box?

Answer: _____ cm^3

3. Pick the best unit for each measurement. Write cm, m^2 , mL, or cm^3 .

(a) the size of a classroom floor _____

(b) the amount of juice in a glass _____

(c) the length of a pencil _____

4. A cereal box is 5 cm long, 3 cm wide, and 2 cm tall. What is its volume?

Answer: _____ cm^3

5. Tomas measures a jewelry box: 6 cm long, 2 cm wide, 3 cm tall. He writes: “Volume $= 6 \times 2 \times 3 = 36 \text{ cm}^2$.”

- (a) Name the error in Tomas’s answer.

Answer: _____

- (b) Write the corrected answer, with its unit.

Answer: _____ cm^3

6. An aquarium is 40 cm long, 20 cm wide, and 30 cm deep. How much water can it hold, in cubic centimeters?

Answer: _____ cm^3

7. Lena's poster board has an area of 2 m^2 . She remembers that each meter is 100 centimeters, so she writes: " $2 \text{ m}^2 = 2 \times 100 = 200 \text{ cm}^2$."

- (a) Name the error in Lena's conversion.

Answer: _____

- (b) Write the corrected number of square centimeters.

Answer: _____ cm^2

8. Challenge. A cube has edges 2 cm long. A second cube has edges twice as long: 4 cm.

- (a) Volume of the small cube:

Answer: _____ cm^3

- (b) Volume of the large cube:

Answer: _____ cm^3

- (c) The large volume is how many times the small volume?

Answer: _____

- (d) Roy says: "The edge doubled, so the volume doubled too." Name the error in Roy's thinking.

Answer: _____